

HACKATHON PROJECT — DETAILED REPORT

AI-Powered Analytics Platform for Karnataka Police

KSP Intelligence — Role-Based Decision Support Platform

1. Executive Summary

This report documents the complete design, build, and current state of KSP Intelligence — an AI-powered analytics platform built for a Karnataka Police hackathon. The platform unifies fragmented FIR records into a single searchable database, applies classical NLP and statistical algorithms to summarize case narratives and detect patterns, and layers a full 10-tier role-based access control system on top, mirroring real police rank hierarchy from Police Constable to DGP.

All core MVP features — ingestion, dashboard, hotspot mapping, AI summarization, similarity matching, and explainable alerting — are complete and demo-ready. Beyond the original MVP scope, the team additionally built a live FIR intake form, a full rank-based login and permission system, and an officer self-service profile page with a secure multi-step password change flow. The email-based OTP delivery step for password changes is the one component still being debugged at the time of this report.

2. Feature Implementation Status

Feature	Status
Data pipeline (ingest, clean, unify)	Done
Backend API & dashboard	Done
Hotspot map	Done
AI case summarizer	Done
Similar-case matching	Done
Explainable alerts	Done
Demo polish (loading/empty/error states)	Done
File New FIR (live intake)	Done
10-tier RBAC login system	Done
Officer profile & edit details	Done
Password change (password + email confirm + OTP)	In progress
Email OTP delivery (Gmail SMTP)	Debugging

3. Core Platform Features

Data Pipeline

- Synthetic, anonymized FIR sample data (35+ records) spanning 6 police units around Bengaluru.
- CSV/JSON ingestion into SQLite with deduplication and schema normalization via `seed_db.py`.
- Unified CaseMaster-based schema mirroring the production FIR ER design (Complainant, Victim, Accused, ArrestSurrender).

Dashboard & Search

- Executive dashboard with category/time-series charts and station/date/category filters.
- Case search by ID, person, vehicle, or location, with a proper “No Records Found” empty state.
- Case detail view with a “Case Not Found” state for unmatched lookups.

Spatial Analysis Map

- Leaflet map centered on Bengaluru plotting all cases by coordinate.
- Grid-based hotspot aggregation highlighting high-concentration zones.

AI Case Summarizer

- Fully offline extractive summarization (TF-IDF sentence scoring) — no external API dependency.
- Summaries cached per case, regenerable on demand, labeled as AI-generated in the UI.

Similar-Case Matching

- Weighted scoring across text similarity, category match, and spatial-temporal proximity.
- Every match returns a plain-language “matched on” explanation, not a bare score.

Explainable Alerts

- Cluster alerts and hotspot-spike alerts generated automatically with human-readable reasons.
- Strictly decision-support — reviewable and dismissible by a human, never auto-actioned.

Live FIR Intake

- “File New FIR” form lets an authorized officer submit a new case through the UI.
- New submissions immediately re-run similarity matching and alert generation, demonstrating the full pipeline on fresh input in real time.

4. Role-Based Access Control (RBAC)

A 10-tier rank hierarchy was implemented on top of the existing Employee/Rank schema, with login IDs following the convention `<RANK>_<ID>_<BATCH>` (e.g. `PC_10452_2015`). Rank is derived from the ID prefix and cross-validated against the database record. Each rank has an explicit, distinct permission set:

Rank	Permitted	Restricted
Police Constable	View assigned FIRs, update patrol status, upload evidence	Register FIR, approve investigations, close cases
Head Constable	Station diary, assist FIR documentation, property records	Approve chargesheets, modify others' investigations
Assistant Sub-Inspector	Update investigation progress, statements, evidence	Approve FIR, finalize investigation, district admin
Sub-Inspector	Register FIR, assign investigation, arrest records	View confidential CID cases, other districts
Inspector / SHO	Full station admin, approve	State-level intelligence, other

	FIRs/updates, assign officers	stations' records
DySP / ACP	Monitor stations, review investigations, approve transfers	State-wide administration
SP / DCP	District dashboard, officer management, approvals	State-level policy admin
IGP	Regional monitoring, regional analytics	State-wide configuration
ADGP	Department-wide monitoring, strategic dashboards	Global system admin
DGP	State-wide access, all districts, policy reports	Cannot alter audit logs without separate controls

5. Officer Profile & Account Security

Every officer, regardless of rank, has a self-service profile page showing their Rank and Login ID as read-only fields (locked to prevent privilege escalation) alongside editable contact details (name, phone, email).

Password changes follow a 4-step secure flow: (1) verify current password, (2) re-confirm the officer's own registered email address, (3) enter a one-time OTP sent to that email, (4) set a new password. On success, all existing sessions are invalidated, forcing re-login with the new credentials. Email delivery uses free Gmail SMTP rather than a paid SMS gateway, keeping the feature cost-free for hackathon use.

Current status: the password-verification and OTP-validation logic (expiry, attempt limits, token issuance) is complete and tested. Real email delivery via Gmail SMTP is being actively debugged — the most likely causes under investigation are missing/incorrect environment credentials, an unverified Gmail App Password, or messages being filtered by the receiving mail server.

6. Technology Stack

Frontend

Component	Choice	Purpose
Framework	React (Vite)	UI framework, fast dev builds
Styling	Tailwind CSS	Utility-first styling
Routing	React Router	Page navigation, active-tab state
Maps	Leaflet.js (react-leaflet)	Hotspot map, incident markers
Charts	Recharts	Dashboard bar/line charts
State	React useState / useEffect / Context	Component state, auth context
HTTP client	Native fetch API	All backend calls

Backend

Component	Choice	Purpose
Framework	FastAPI (Python)	REST API server
ORM	SQLAlchemy	Python-to-SQL mapping
Database	SQLite	MVP data store
Auth	JWT tokens + passlib/bcrypt	Login sessions, password hashing
Email	smtplib + Gmail SMTP	OTP delivery
Data loading	pandas (seed_db.py)	CSV ingestion

Security

Component	Choice	Purpose
Password hashing	bcrypt via passlib	Never stores plaintext
Session tokens	Signed JWT	Carries EmployeeID + RankID
RBAC	Custom permissions.py allow-lists	10-tier rank hierarchy
OTP	6-digit, 5-min expiry, 5-attempt limit	Password-change verification

7. Algorithms Used

Deliberately classical, deterministic algorithms were chosen over trained ML models or LLM APIs — this keeps the platform fully offline (no demo-day network dependency) and keeps every AI output explainable, directly satisfying the governance principle from the original research that predictive outputs must never be a black box.

Component	Choice	Purpose
TF-IDF	Summarizer, similarity matcher	Scores word/sentence distinctiveness
Extractive summarization	nlp_summarizer.py	Picks top-scoring sentences from BriefFacts
Cosine similarity	similarity_matcher.py	Measures closeness of two TF-IDF vectors
Haversine formula	similarity_matcher.py, hotspots	Real-world GPS distance
Grid-based spatial binning	/analytics/hotspots	Groups cases into lat/lon cells
Weighted linear scoring	find_similar_cases()	0.5 text + 0.3 category + 0.2 spatial
Rule-based threshold alerting	alert_engine.py	Fires alert when count exceeds threshold

8. Design Principles

- Human review before any operational action on AI outputs — alerts are surfaced, never auto-executed.
- Explainable reasons attached to every alert and case match (the “matched on” field).
- Rank-appropriate access — no officer can see or do more than their role permits.
- Synthetic, anonymized data only — no real case data used at any stage.
- Identity fields (Rank, Login ID) are locked from self-editing to prevent privilege escalation.

9. Known Issues / In Progress

- Email OTP delivery via Gmail SMTP is not yet reliably reaching inboxes — under active debugging (App Password validity, sender/recipient filtering, and DEMO_MODE flag state are the leading suspects).
- Password change flow is functionally complete except for this final delivery step.

10. What This Demonstrates

Beyond the original MVP scope, this build additionally proves a production-relevant concern the original report flagged directly: that predictive/AI-assisted policing tools introduce governance risk without proper access control. The RBAC layer, locked identity fields, and audit-conscious password-change flow show the platform was extended specifically to address that risk, not just the core intelligence features.

11. Conclusion

The platform delivers the complete narrative set out in the original proposal — unified records, AI-generated summaries and case linkages, spatial pattern detection, and explainable, human-reviewed alerts — and extends it with a realistic rank-based access model and officer account security. With the OTP delivery issue resolved, the platform will be fully feature-complete and demo-ready end to end.